

Ahmed Attia

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Location: Masdar City, Abu Dhabi, UAE

EDUCATION

- **Mohamed bin Zayed University of Artificial Intelligence** Abu Dhabi, United Arab Emirates
Master of Science in Natural Language Processing; GPA: 3.7/4.0 Aug 2024 - Present
- **Kafr El-Sheikh University Faculty of Artificial Intelligence** Kafr El-Sheikh, Egypt
Bachelor of Science in Artificial Intelligence; GPA: 3.6/4.0 Aug 2020 - Jul 2024

EXPERIENCE

- **R&D Machine Learning Engineer at Yalla Group Limited** Dubai, UAE
Full-time May 2025 - July 2025
 - Developed an AI agent to play the classical board game Draw Dominoes, leveraging deep reinforcement learning.
 - Implemented the agent using the Neural Fictitious Self-Play (NFSP) algorithm, achieving a 72.5% win rate against a uniform random baseline—1.5% higher than the previous benchmark.
 - Designed and deployed a distributed Actor-Learner architecture for scalable training using **Ray**, enabling 24 parallel actors for experience generation and centralized gradient aggregation at the learner.
 - The whole system was deployed on Google Kuburnetes Engine (GKE) on GCP.
- **Machine Learning Engineer at SpeedLegal** California, USA
Full-time Jul 2022 - Jun 2024
 - Collaborated in a team to build a Minimum Viable Product in 2 months for the goal of automating contract review process to help lawyers extract salient portions from contracts easily.
 - Developed Retrieval Augmented Generation pipelines with LLMs (**GPT-4**). Thereafter, properly evaluating the information retrieval system using precision@k, recall@k & mean reciprocal rank. This improved the contract review process by iterating and reaching a high recall@6 of 96% for NDA agreements.
 - Fine-tuned a pre-trained **BERT** model for an extractive question answering task on license agreements. The outcome was that the extractive model was able to do average of 20% exact match score and 65% on a set of 16 clauses leading to approximately 15 minutes decrease in contract review by the legal team.
 - Explored model Interpretability and Explainability (**LIME**). This helped in uncovering the details of the model reasoning in saying why it classified certain legal text under certain classes for reliability.

SKILLS SUMMARY

- **Languages and Frameworks:**
C, Python, MongoDB, Unix scripting, Numpy, Pandas, Scikit-learn, PyTorch, Redis, Docker, Ray, Google Cloud Platform (GCP), GitLab CI, Streamlit, Design Patterns, Version Control, RESTful API, SOLID principles, Object Oriented Programming, Operating Systems, Agile Software Development, Reinforcement learning, Information retrieval.
- **Languages:**
Arabic: Native, English: Fluent

PUBLICATIONS

- **Accurate and Diverse LLM Mathematical Reasoning via Automated PRM-Guided GFlowNets:**
Adam Younsi*, **Ahmed Attia***, Abdalgader Abubaker, Mohamed El Amine Seddik, Hakim Hacid, Salem Lahlou. ArXiv

RESEARCH PROJECTS

- **DNG-MCTS— Reinforcement learning, Pytorch, Numpy, Scipy** MBZUAI Nov 2024
 - Implemented the paper: Bayesian mixture modelling and inference based Thompson sampling in Monte-Carlo tree search on **RaceTrack** environment.
 - Analyzed the performance of the DNG-MCTS variant against UCT MCTS, demonstrating faster convergence in **RaceTrack**.
 - Enhanced the algorithm's scalability and adaptability for TikZ program generation, integrating it with LLMs using the DeTikZify Vision-Language Model during inference.

REFERENCES

Available upon request.